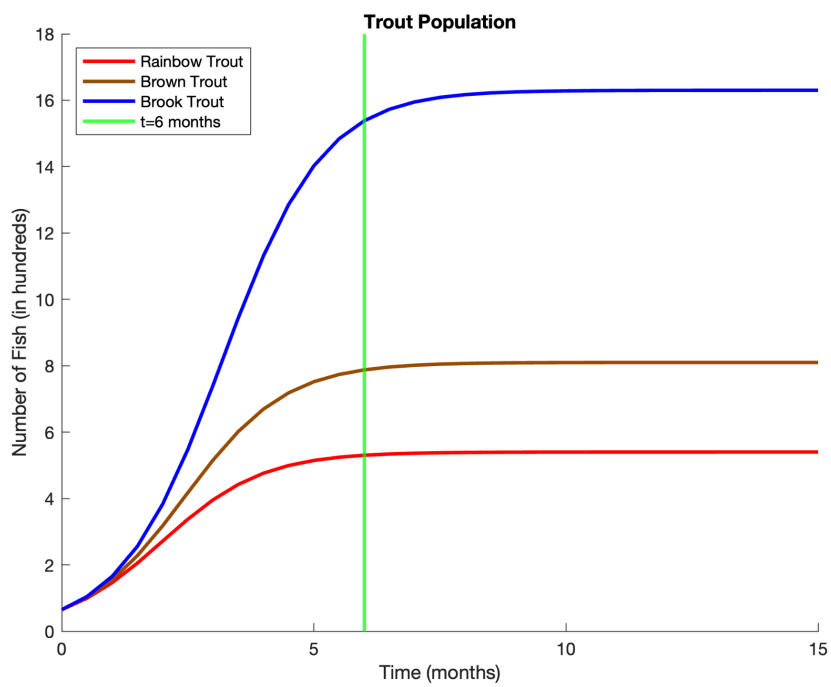
## I Introduction

We live in a rural fishing town and would like to determine the most plentiful species of trout to stock one of our lakes with. By gaining a better understanding of how the population of certain fish species in the lake changes over time, we can determine which species of trout will be the most successful over time with population harvesting taking place. We can do this by first examining the equation $\frac{dy}{dt} = r(1 - \frac{y}{L})y - h(y)$. (Equation 1) Here, y(t) represents the change of fish population over time t (in months), r represents the intrinsic growth rate and L is the carrying capacity. y is calculated in units of hundreds of fish. h(y) in equation 1 is representative of the equation $h(y) = \frac{py^2}{q+y^2}$. (Equation 2). h(y) simply represents the harvesting function, where p and q are positive real numbers representing how effective the fishermen are. Throughout this report, we will break down each element of Equation 1 to understand which species of fish will thrive in a rural fishing environment.

## II Population Analysis

The first step in understanding how the fish population is affected by harvesting is to better understand the growth of each species' population. To do this, we will inspect the logistic growth of each species over time. This can be done by solving equation 1. The solution is y(t) = $\frac{L}{1+(\frac{L}{y_0}-1)e^{-rt}}$ where y(0) = $y_0$. Each species of fish has a unique carrying capacity, *L*, or the maximum population size without any exterior factors to limit growth. Brook trout have a maximum population of 1,630 fish *(L = 16.3)*, brown trout have a maximum population of 810 fish *(L = 8.1)*, and rainbow trout have a maximum population of 540 fish *(L=5.4)*. Each standard

logistic model without harvesting has been plotted below to neatly compare the growth of each species population over 15 months, assuming the initial population of each species is 100 fish.




Visualization of Logistic Population Growth

*Figure 1*

Using this graph, we can conclude what the population size of each species, in hundreds, will be at t = 6 , or after 6 months of growth. At 6 months time the approximate population of rainbow trout, brown trout, and brook trout are respectively 400, 520, and 740 fish.

Now that we have an understanding of each species population growth alone, we can introduce the exterior factor of harvesting. Using the harvesting function, equation 2, we can

visually evaluate how each species is affected by harvesting. Each population of fish was evaluated at 3 values of p and q to highlight differing fishermens successes.

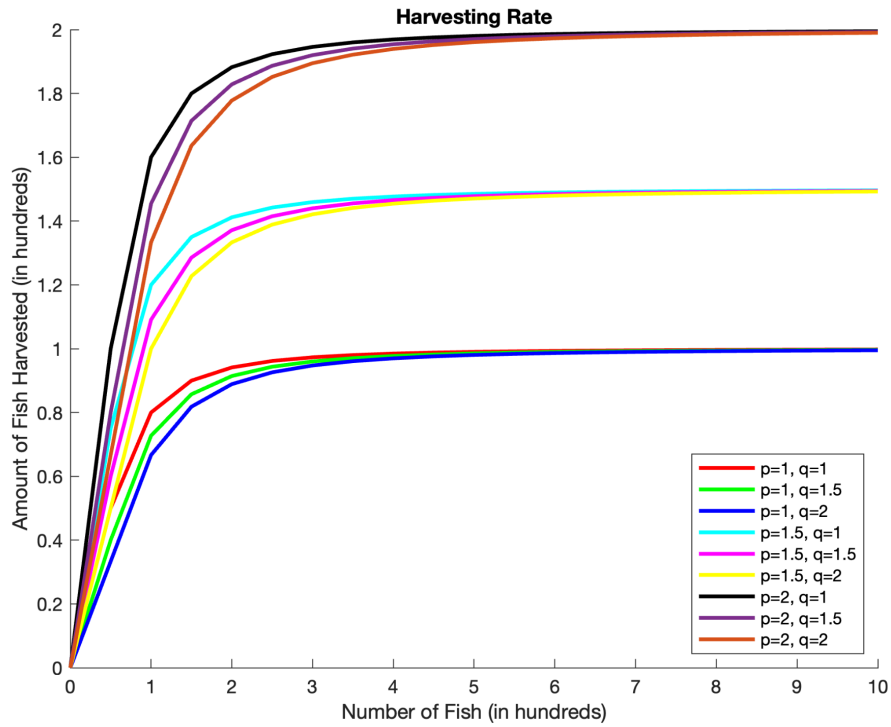



*Figure 2*

We can see in Figure 2 that, as y increases, h(y) approaches the constant value of p. This shows that a sufficient population has been established for both successful fishing as well as consistent long term fish population. Note that adding to the fish population will not significantly increase h(y) because the numerator $py^2$ and the denominator $q+y^2$ are growing at similar rates. When our population of fish is close to zero, the amount of fish harvested is also close to zero; this makes sense since there is a limited amount of fish for fishermen to catch.

## III Equilibrium Solutions

An equilibrium solution is where the rate of change of a differential equation is zero at all values or, simply put, the zeros of f (y) . In our case, this is where the fishing rate and the population growth rate are both zero. Each fish species has a different carrying capacity (*L*), or the maximum population size each species could sustain indefinitely. Below, the equation $f(y) = r(1 - \frac{y}{L})y - \frac{py^2}{q+y^2}$ is graphed, where *f(y) = y'(t).*

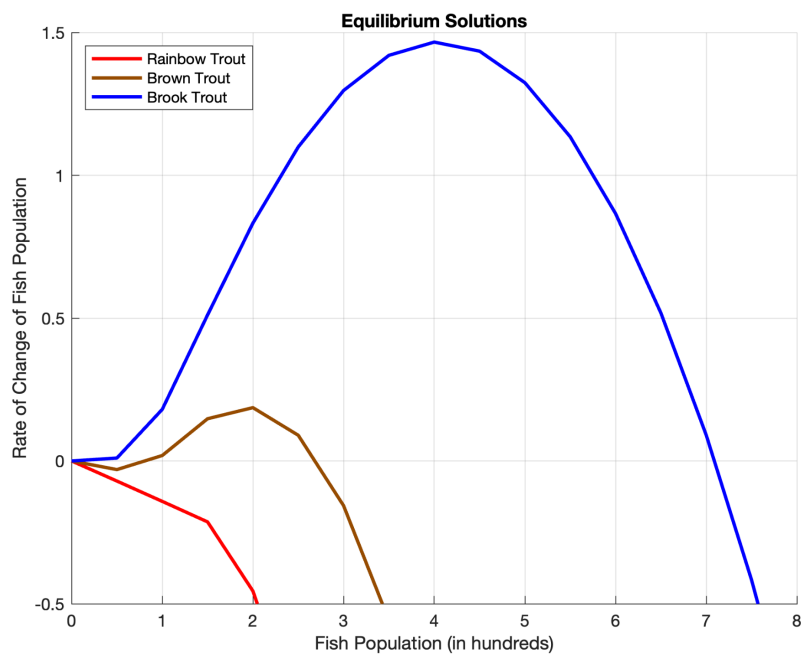



*Figure 3*

Starting by evaluating the rainbow trout with a carrying capacity of 540 fish (*L=5.4*), we can identify the equilibrium points that take place at approximately y = 0 and y = 0.7, or 0 and 70 fish. Moving onto the brown trout with a carrying capacity of 810 fish (*L=8.1)* we can identify the equilibrium points are at approximately y = 0, y = 0.8, y = 1.9, and y = 5.4; or 0, 80, 190 and 540 fish in the lake. Finally, examining the brook trout with a carrying capacity of 1630 (*L=16.3*)

we can identify the equilibrium points that take place at approximately y = 0 and y = 14.2, or 0 and 1420 fish.

Using MATLAB we can more accurately determine where each species' equilibrium points lie. After doing this we obtain the following values in hundreds of fish; y = 0, y = 0.7050 for rainbow trout; y = 0, y = 0.8018, y = 1.8564, y = 5.4418 for brown trout; and y = 0, y = 14.1898 for brook trout.

Using these equilibrium values, we can plot each value on the corresponding direction fields using the logistic model for growth of each fish population. The direction fields were plotted using the provided dirfield.m file altered to include the equilibrium solutions on the graph.

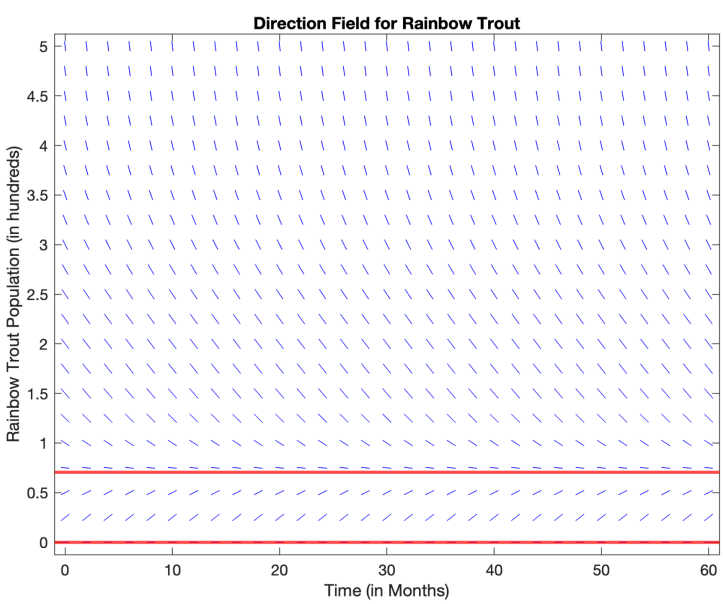



*Figure 4*

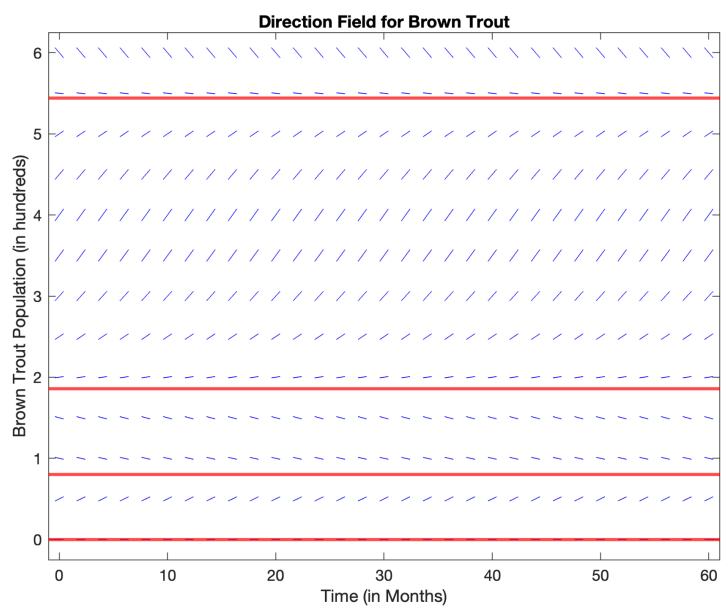



*Figure 5*

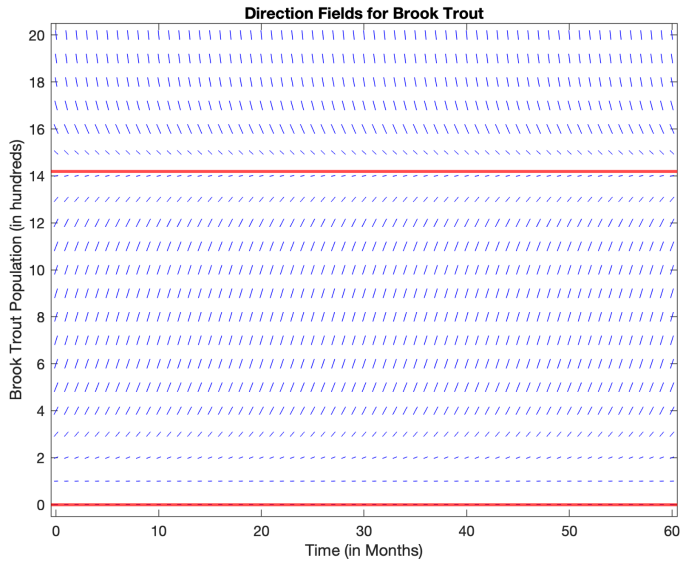



*Figure 6*

Figure 4 depicts that, as time passes, the Rainbow Trout population will decrease to 70 fish if the initial population is above 70 fish. If the initial population is below 70 fish, the population will increase before holding steady at 70 fish. This trend of approaching 70 fish from either direction indicates that the equilibrium solution of 0.07050 (approximately 70 fish) is a stable solution to the logistic population model for Rainbow Trout.

Figure 5 depicts the three non-zero, and one zero, equilibrium solutions for the Brown Trout population. If the initial population were above 544 fish, we would see a steady decrease in the population approaching 544. Similarly, if the initial population was lower than 186 fish but higher than 80 fish, we would see a steady decrease approaching 80. Conversely, if the initial population were to be less than 80 but greater than 0, we would see an increase until the population reached 80. If the initial population was to be larger than 186 but lower than 544, we would see an increase in population approaching 544. From this data, we can determine that

5.4418 (544 fish), 1.8564 (186 fish), and 0.8018 (80 fish) are all stable equilibrium solutions of the logistic population model for Brown Trout.

Figure 6 depicts the two equilibrium solutions for Brook Trout. From the graph we can discern that, given that the initial population is greater than 0, the Brook Trout population will always approach 14.1898 (1,419 fish). Since the population approaches this value from either direction, 14.1898 is a stable equilibrium solution to the logistic population model for Brook Trout.

Since the value of y = 0 is unstable in all three species cases, we can determine that the population of any of the fish species will never reach zero, since none of the solution curves ever approach 0 as t approaches ∞.

## IV Results

In order to best visualize the long term behavior of each species of fish over a longer period of time, we employed Euler's approximation to solve the logistic equation representing fish population with harvesting. Euler's Approximation is as follows $y_{i+1} = y_i + hf(t_i, y_i)$. Each approximation runs from zero months to 60 months time, and was run 4 times to visualize the effect of different initial population values on the population growth.

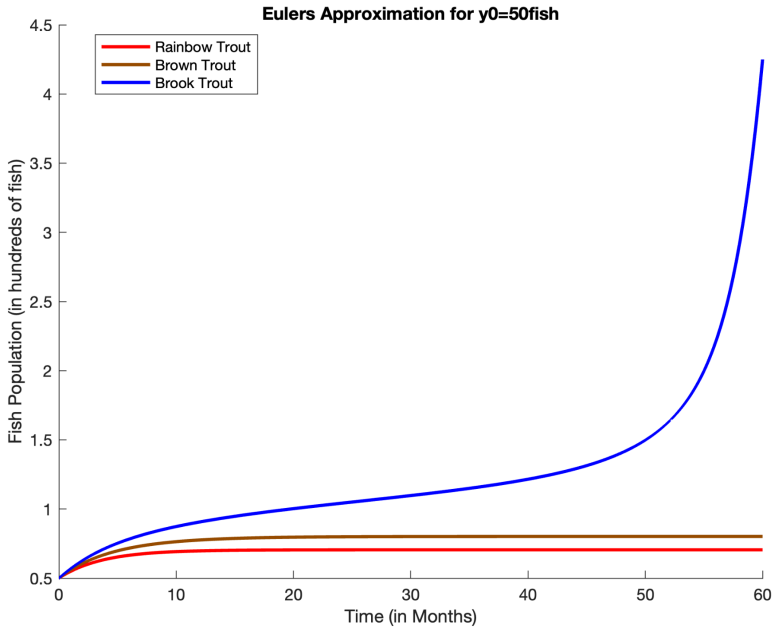



*Figure 7*

The above graph shows how each population of fish will change after 60 months of harvesting with an initial 50 fish in the lake. We can see that both the brown trout and the rainbow trout will not respond well to fishing in the lake. Their populations initially increase slightly, but do not increase over time, meaning fishing locals may grow unhappy. The brook trout species is most likely to thrive in this specific lake situation.

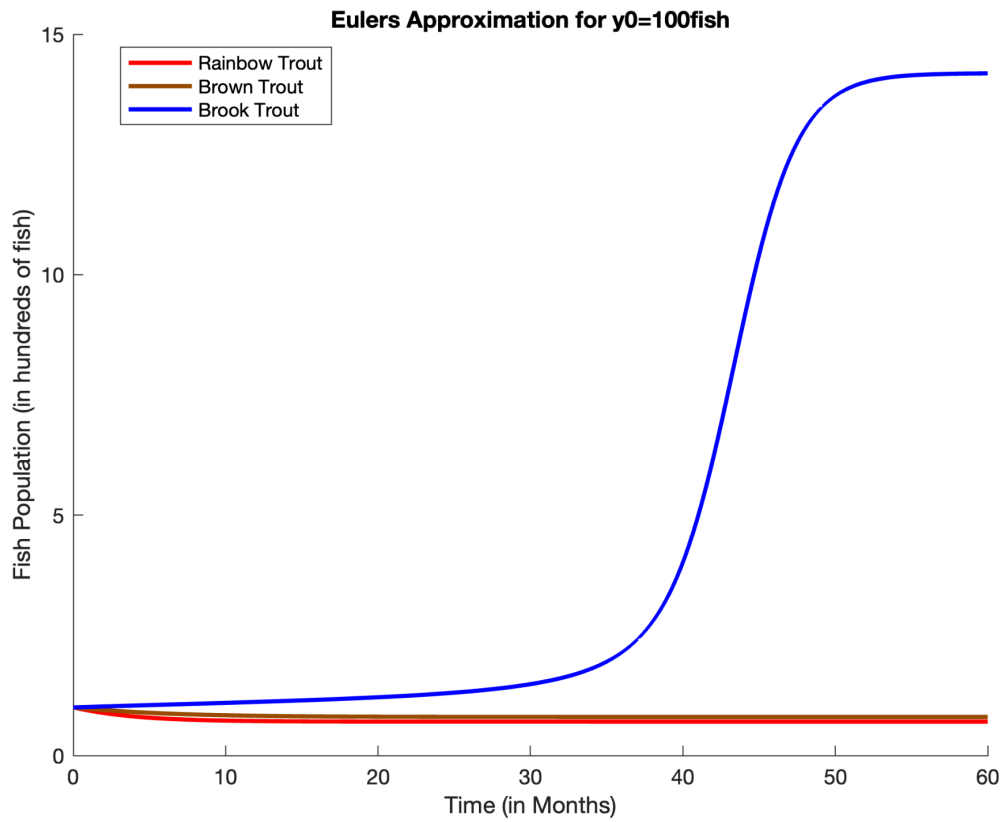



*Figure 8*

In this approximation model, each fish population at time zero is 100 fish. We can see similar results as the first approximation we made with t=0 at 50 fish. Both the brown and rainbow trout species don't thrive nearly as well as the brook trout population. In this model though, we can see that, around 50 months, the brook trout population plateaus. This plateau, or lack of a rate of change, takes place at a much higher population than the other two species, likely making it the best to stock the late and result in happy fishermen.

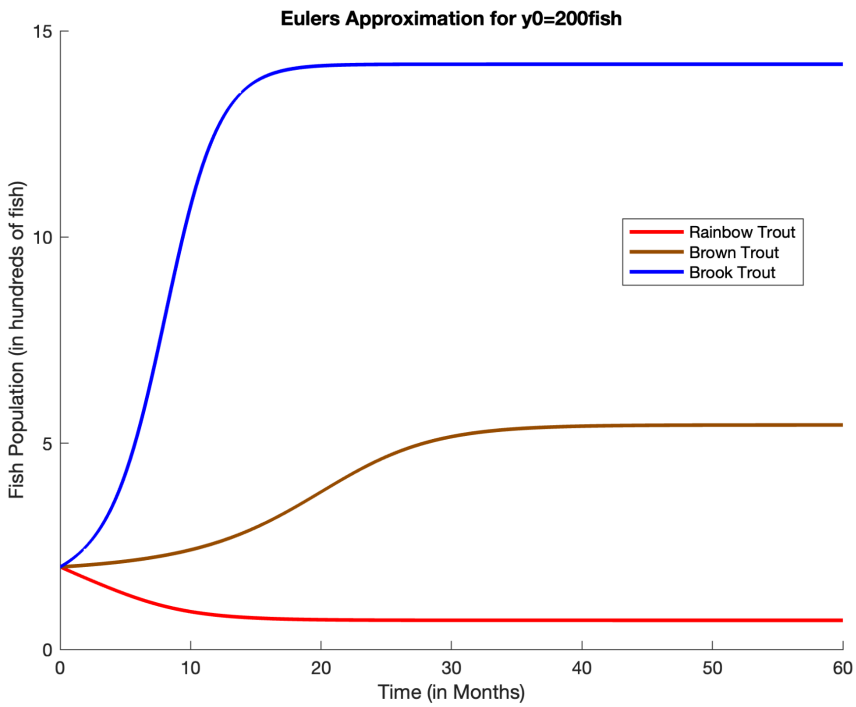



*Figure 9*

For this approximation, each species of trout begins with a population of 200 fish. We can see that the behavior of the rainbow trout is similar to that of the previous two models. The population plateaus around 10 months at a population of less than 150 fish. This is less than ideal for a popular fishing spot in a rural town. Opposed to the rainbow trout population, the brown trout population does slightly better in this model; initially increasing and plateauing around 30 months. But, yet again, the brook trout population thrives. The population grows significantly in the first 15 months, and plateaus at a population of approximately 1,400 (notice this is the equilibrium value from before). Yet again, brook trout would be the best species to stock the late with.

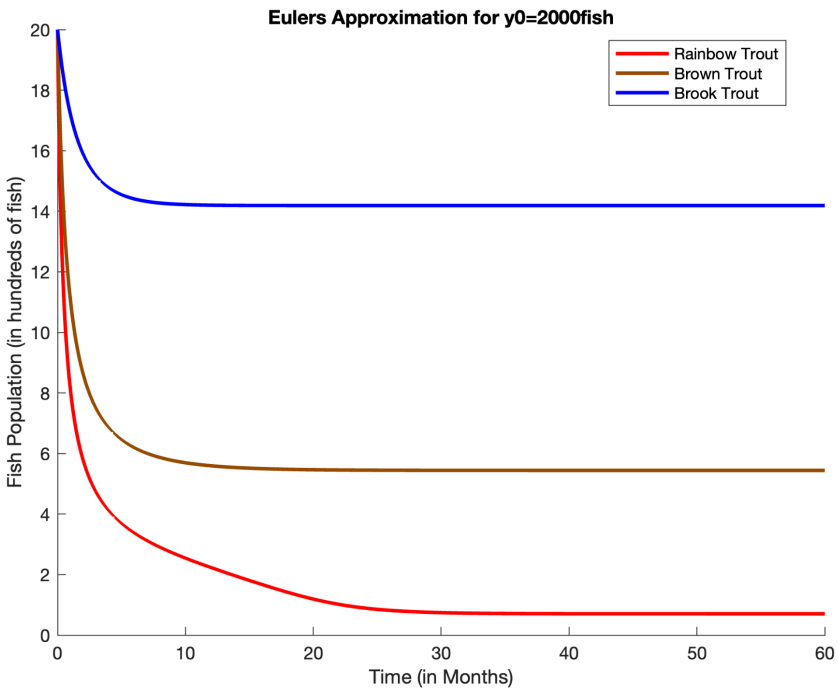



*Figure 10*

In this final approximation, we set the initial population of each fish species to 2,000. Compared to the other models, each fish population in the first 10 months drastically decreases, with a large rate of change over the first 10 months. It takes each population less time to reach their equilibrium values or plateau. Yet again though, we can see that the brook trout population remains the highest over time with consistent fishing taking place in the lake.

## IV Conclusion

Overall, based on the research conducted on each of the three fish populations evaluated with assistance from MATLAB, the best species to stock the lake with is brook trout. The population over 60 months stabilizes to approximately 1,400 fish total, allowing for a sufficient amount of fish to be harvested without significantly harming the trout population and allowing for satisfied fishermen.

It is important to consider that these models are simplified and, therefore, have a slight margin of error. Other outside factors will affect the fish populations such as water quality, predatory species, and natural disasters, as well as the species competing. While these may not significantly affect the error of our approximations, they are important to consider on a broader scope as they could affect the long term population of fish in the lake.